Notes on AR(1) Time Series Dynamics: Stationarity, Subsampling Thinning, Autocorrelation, and Entropy Estimation

Research Notes

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Part A: Defining the Model and Stationarity Condition

We define the first-order autoregressive process, AR(1), for $t \in \mathbb{Z}$ as:

$$X_t = C + \phi_1 X_{t-1} + \epsilon_t \tag{1}$$

where:

- X_t is the value of the time series at discrete time index t.
- C is the constant intercept term.
- ϕ_1 is the autoregressive coefficient representing the persistence or "memory" of past shocks ($|\phi_1| < 1$ for stationarity; $\phi_1 \to 0$ corresponds to independent noise, while $\phi_1 \to 1$ indicates a highly persistent process).
- $\epsilon_t \sim \mathcal{N}(0, s^2)$ is a Gaussian white noise innovation process, independent of X_{t-1}.

Without loss of generality, setting $C = 0$ shifts the distribution mean to zero without altering the differential entropy or autocovariance structure of the process.

Stationary Variance Derivation

Let $\text{Var}(X_{t-1}) = \sigma^2$ and $\text{Var}(\epsilon_t) = s^2$. Using the linear variance property $\text{Var}(aX) = a^2\text{Var}(X)$, we have:

$$\text{Var}(\phi_1 X_{t-1}) = \phi_1^2 \sigma^2 \tag{2}$$

Since ϵ_t is strictly independent of X_{t-1}, the variance of the sum is additive:

$$\text{Var}(X_t) = \phi_1^2 \sigma^2 + s^2 \tag{3}$$

Enforcing the stationarity requirement $\text{Var}(X_t) = \text{Var}(X_{t-1}) = \sigma^2$:

$$\sigma^2 = \phi_1^2 \sigma^2 + s^2 \implies \sigma^2(1 - \phi_1^2) = s^2 \tag{4}$$

$$s^2 = \sigma^2(1 - \phi_1^2) \tag{5}$$

Limiting Cases Check

- As $\phi_1 \to 0$, $s^2 \to \sigma^2$: the process reduces to pure independent noise.
- As $\phi_1 \to 1$, $s^2 \to 0$: the process becomes highly persistent, changing very little between consecutive time steps.

Part B: Marginal Distribution Stationarity

We show by induction on t that the marginal distribution of X_t remains $\mathcal{N}(0, \sigma^2)$ for all $t \geq 0$.

Proof by Induction

- **Base Case ($t = 0$):** $X_0 \sim \mathcal{N}(0, \sigma^2)$.
  Note: We initialize X_0 from $\mathcal{N}(0, \sigma^2)$ rather than a deterministic value like 0 to ensure the stationary marginal distribution holds immediately from $t = 0$ without burn-in transients.
- **Inductive Step:** Assume $X_{t-1} \sim \mathcal{N}(0, \sigma^2)$. Then:

$$X_t = \phi_1 X_{t-1} + \epsilon_t \tag{6}$$

  Since X_t is a linear combination of two independent normal random variables ($\phi_1 X_{t-1}$ and ϵ_t), X_t is itself Gaussian.
- **Mean:**

$$\mathbb{E}[X_t] = \phi_1 \mathbb{E}[X_{t-1}] + \mathbb{E}[\epsilon_t] = \phi_1(0) + 0 = 0 \tag{7}$$

- **Variance:**

$$\mathsf{Var}(X_t) = \phi_1^2 \sigma^2 + s^2 = \phi_1^2 \sigma^2 + \sigma^2(1 - \phi_1^2) = \sigma^2 \tag{8}$$

Thus, $X_t \sim \mathcal{N}(0, \sigma^2)$ for all $t \geq 0$.

Part C: Differential Entropy of the Marginal Distribution

For a continuous univariate Gaussian random variable $X \sim \mathcal{N}(0, \sigma^2)$, the probability density function $f_X(v)$ is:

$$f_X(v) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{v^2}{2\sigma^2}\right), \quad v \in \mathbb{R} \tag{9}$$

Taking the natural logarithm:

$$\ln f_X(v) = -\frac{1}{2}\ln(2\pi\sigma^2) - \frac{v^2}{2\sigma^2} \tag{10}$$

The differential entropy $h(X)$ is defined as:

$$h(X) = -\mathbb{E}[\ln f_X(X)] = -\mathbb{E}\left[-\frac{1}{2}\ln(2\pi\sigma^2) - \frac{X^2}{2\sigma^2}\right] = \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2\sigma^2}\mathbb{E}[X^2] \tag{11}$$

Since $\mathbb{E}[X] = 0$, the second raw moment equals the variance: $\mathbb{E}[X^2] = \mathsf{Var}(X) = \sigma^2$. Substituting this gives:

$$h(X) = \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2\sigma^2}(\sigma^2) = \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2} \tag{12}$$

Rewriting $\frac{1}{2} = \frac{1}{2}\ln(e)$ and applying logarithmic identities yields:

$$h(X) = \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2}\ln(e) = \frac{1}{2}\ln(2\pi e\sigma^2) \tag{13}$$

Key Remark: The intercept C does not appear in $h(X)$ because differential entropy is shift-invariant and measures uncertainty strictly relative to the dispersion around the mean.

Part D: Subsampling / Thinning Proof (Recording Every k-th Value)

Let X_t be the stationary AR(1) process defined above. Consider the thinned sequence formed by retaining every k-th observation:

$$Y_j = X_{jk}, \quad j \in \mathbb{N},\ k \in \mathbb{Z}^+ \tag{14}$$

We claim that Y_j is itself a stationary AR(1) process with AR parameter $\psi = \phi_1^k$, innovation variance $\mathsf{Var}(W_j) = \sigma^2(1 - \phi_1^{2k})$, and identical marginal distribution $\mathcal{N}(0, \sigma^2)$.

1. Unrolling the Recursion

By repeated substitution of $X_t = \phi_1 X_{t-1} + \epsilon_t$:

$$k = 1: \quad X_{t+1} = \phi_1 X_t + \epsilon_{t+1} \tag{15}$$

$$k = 2: \quad X_{t+2} = \phi_1(\phi_1 X_t + \epsilon_{t+1}) + \epsilon_{t+2} = \phi_1^2 X_t + \phi_1 \epsilon_{t+1} + \epsilon_{t+2} \tag{16}$$

$$k = 3: \quad X_{t+3} = \phi_1^3 X_t + \phi_1^2 \epsilon_{t+1} + \phi_1 \epsilon_{t+2} + \epsilon_{t+3} \tag{17}$$

By induction, for any step size $k \geq 1$:

$$X_{t+k} = \phi_1^k X_t + \sum_{i=0}^{k-1} \phi_1^i \epsilon_{t+k-i} \tag{18}$$

Substituting $t = (j-1)k$, we obtain $X_{t+k} = X_{jk} = Y_j$ and $X_t = X_{(j-1)k} = Y_{j-1}$:

$$Y_j = \phi_1^k Y_{j-1} + W_j, \quad \text{where } W_j = \sum_{i=0}^{k-1} \phi_1^i \epsilon_{(j-1)k+k-i} \tag{19}$$

2. Distribution and Variance of Innovation Sum W_j

Since W_j is a linear combination of independent zero-mean normal variables $\epsilon_\tau \sim \mathcal{N}(0, s^2)$, W_j is normally distributed with mean zero:

$$\mathbb{E}[W_j] = \sum_{i=0}^{k-1} \phi_1^i \mathbb{E}[\epsilon_{(j-1)k+k-i}] = 0 \tag{20}$$

Using independence across time steps, the variance of the sum is the sum of the variances:

$$\mathsf{Var}(W_j) = \sum_{i=0}^{k-1} (\phi_1^i)^2 \mathsf{Var}(\epsilon) = s^2 \sum_{i=0}^{k-1} (\phi_1^2)^i \tag{21}$$

Applying the finite geometric series formula $\sum_{i=0}^{k-1} r^i = \frac{1-r^k}{1-r}$ with $r = \phi_1^2$:

$$\mathsf{Var}(W_j) = s^2 \frac{1 - (\phi_1^2)^k}{1 - \phi_1^2} \tag{22}$$

Substituting $s^2 = \sigma^2(1 - \phi_1^2)$:

$$\mathsf{Var}(W_j) = \sigma^2(1 - \phi_1^2) \cdot \frac{1 - \phi_1^{2k}}{1 - \phi_1^2} = \sigma^2(1 - \phi_1^{2k}) \tag{23}$$

3. Independence Verification

The thinned innovation W_j depends exclusively on future noise terms relative to time $(j-1)k$, namely $\{\epsilon_{(j-1)k+1}, \ldots, \epsilon_{jk}\}$. Conversely, $Y_{j-1} = X_{(j-1)k}$ depends only on initial state X_0 and past noise terms $\{\epsilon_1, \ldots, \epsilon_{(j-1)k}\}$.

Because all underlying noise terms are mutually independent, W_j is strictly independent of Y_{j-1}. Thus, Y_j is a valid $\mathsf{AR}(1)$ process with autoregressive parameter $\psi = \phi_1^k$.

Part E: Autocovariance Structure and Effective Sample Size

1. Autocovariance Structure of X_t

For lag $m \geq 0$:

$$\gamma(m) = \mathsf{Cov}(X_t, X_{t+m}) = \mathsf{Cov}\left(X_t,\ \phi_1^m X_t + \sum_{i=0}^{m-1} \phi_1^i \epsilon_{t+m-i}\right) \tag{24}$$

Since X_t is independent of all future innovations ϵ_{t+m-i} ($i < m$):

$$\gamma(m) = \phi_1^m \mathsf{Var}(X_t) = \sigma^2 \phi_1^m \implies \rho(m) = \frac{\gamma(m)}{\gamma(0)} = \phi_1^m \tag{25}$$

2. Integrated Autocorrelation Time (τ_{int}) and Effective Sample Size (N_{eff})

The integrated autocorrelation time quantifies how many dependent steps equal one independent observation:

$$\tau_{\mathsf{int}} = 1 + 2\sum_{m=1}^{\infty} \rho(m) = 1 + 2\sum_{m=1}^{\infty} \phi_1^m = 1 + 2\left(\frac{\phi_1}{1-\phi_1}\right) = \frac{1+\phi_1}{1-\phi_1} \tag{26}$$

For a thinned series Y_j with step k, the lag-1 autocorrelation is $\phi_k = \phi_1^k$, giving:

$$\tau_{\mathsf{int}}(k) = \frac{1+\phi_1^k}{1-\phi_1^k} \tag{27}$$

For a retained sample of size N, the Effective Sample Size $N_{\mathsf{eff}}(k)$ is:

$$N_{\mathsf{eff}}(k) = \frac{N}{\tau_{\mathsf{int}}(k)} = N\left(\frac{1-\phi_1^k}{1+\phi_1^k}\right) \tag{28}$$

Numerical Illustration

Consider a highly persistent process with $\phi_1 = 0.99$:

- **Without thinning ($k = 1$):** $\tau_{\mathsf{int}} = \frac{1.99}{0.01} = 199 \implies N_{\mathsf{eff}} \approx 0.005N$.
- **With thinning ($k = 20$):** $\phi_{\mathsf{eff}} = 0.99^{20} \approx 0.817 \implies N_{\mathsf{eff}} \approx 0.10N$ per retained sample (a 20-fold increase in independent information content per saved observation).

Part F: Marginal Differential Entropy vs. Process Entropy Rate

1. Joint Entropy of N Consecutive Observations

By the chain rule of entropy for $X_N = (X_1, X_2, \ldots, X_N)$:

$$h(X_N) = h(X_1) + \sum_{t=2}^{N} h(X_t \mid X_{t-1}) \tag{29}$$

Since $X_t \mid X_{t-1} \sim \mathcal{N}(\phi_1 X_{t-1}, s^2)$, conditional entropy depends strictly on the noise variance $s^2 = \sigma^2(1-\phi_1^2)$:

$$h(X_t \mid X_{t-1}) = \frac{1}{2}\ln(2\pi e s^2) = \frac{1}{2}\ln\left(2\pi e \sigma^2(1-\phi_1^2)\right) \tag{30}$$

Thus, the exact joint entropy is:

$$h(X_N) = \frac{1}{2}\ln(2\pi e\sigma^2) + \frac{N-1}{2}\ln\left(2\pi e\sigma^2(1-\phi_1^2)\right) \tag{31}$$

2. Process Entropy Rate

The asymptotic entropy rate $h'(X)$ is:

$$h'(X) = \lim_{N\to\infty} \frac{h(X_N)}{N} = \frac{1}{2}\ln\left(2\pi e\sigma^2(1-\phi_1^2)\right) \tag{32}$$

3. Key Distinction for Estimators

The difference between marginal entropy and entropy rate equals the mutual information $I(X_t; X_{t-1})$:

$$\Delta h = h(X_t) - h'(X) = -\frac{1}{2}\ln(1-\phi_1^2) = I(X_t; X_{t-1}) \geq 0 \tag{33}$$

Practical Implication: A 1D nearest-neighbor estimator operating on scalar points y_i measures the *marginal differential entropy* $h(X) = \frac{1}{2}\ln(2\pi e\sigma^2)$. Estimating the *entropy rate* $h'(X)$ requires multi-dimensional delay-coordinate embedding.

Part G: Kozachenko-Leonenko (k-NN) Estimator Reference

For N scalar samples $Y_1, Y_2, \ldots, Y_N$ drawn i.i.d. or stationarily from a continuous density $f(y)$, the Kozachenko-Leonenko differential entropy estimator is given by:

$$\hat{h}_{k\text{NN}}(Y) = \psi(N) - \psi(k_{\text{nn}}) + \ln(2) + \frac{1}{N}\sum_{i=1}^{N}\ln(\rho_i) \tag{34}$$

where:

- k_{nn} is the selected number of nearest neighbors (typically $k_{\text{nn}} \in \{2, 3, 4, 5\}$).
- $\rho_i = |Y_i - Y_{(i;k_{\text{nn}})}|$ is the Euclidean distance from sample point Y_i to its k_{nn}-th nearest neighbor in the dataset.
- $\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}$ is the digamma function. For integer m, $\psi(m) = -\gamma + \sum_{j=1}^{m-1}\frac{1}{j}$, where $\gamma \approx 0.5772156$ is the Euler-Mascheroni constant.